A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 11

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 4 points below. **1** Warming melts ice shelves that hold back glaciers and lubricates ice flow so land ice reaches the sea faster (Antarctic Peninsula most affected) – sea ice loss removes krill habitat and lowers albedo **1** Raised UV harms planktonic algae, krill and fish / sea urchin larvae in surface waters (land animals protected by fur / feathers) **1** Disturbance of wildlife, introduction of non-indigenous species and pathogens, fuel and sewage pollution – visitor numbers rising rapidly at a few landing sites **1** Reserves of coal, oil, gold, uranium, platinum etc – pressure to exploit will rise as reserves elsewhere deplete though currently banned and operations extremely difficult **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **albedo** → reflectivity / the proportion of solar radiation reflected / how much sunlight is bounced back **Do not credit** • 'ice reflects the heat back' - albedo is the reflection of incoming short-wave solar radiation, before it is absorbed	4

Do not write outside the box

Question	Marking guidance	Marks
01.2	**1 mark each** 1 mark for each of the 6 points below. **1** Treaty signed by 52 countries including all active there – no military activity (except supporting research), no nuclear explosions or radioactive waste dumping **1** No commercial mineral exploitation **1 Fisheries regulated by CCAMLR:** krill and some fish exploited, fisheries closed if over-exploited – no commercial exploitation of mammals **1 Tourism control:** no accommodation on land (wastes stay on ships), one ship per site with maximum 100 ashore, trained guides, keep 5 m from wildlife, no litter, cleaned protective clothing to prevent introductions **1 Waste management:** all wastes removed (fuel containers, batteries, plastics, oil) – sewage only discharged where sea currents disperse it – old waste removed unless removal causes more damage **1** Only long-term residents are research scientists and support staff – strict regulation of research impacts **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'nuclear power releases no carbon dioxide' - mining, fuel processing, construction and decommissioning all do	6
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 69-71; AQA spec 3.1.2.3.4.	
02.1	**2 marks per point** 2 marks for each of the 3 points below; award 1 mark where a point is only partly made. **2** Photosynthesis and aerobic respiration balance O2 and CO2 in a dynamic equilibrium – evapotranspiration adds water vapour forming clouds and rain – microbes drive the nitrogen and other biogeochemical cycles so wastes do not accumulate and nutrients are recycled **2** Detritivores, fungi and bacteria decompose dead organic matter releasing nutrients and forming humus that binds soil – organic acids weather rock forming new soil – roots bind soil, reducing erosion and regulating water flow / flooding **2** Species provide food, pollination (bees, moths, birds, bats - more reliable than wind, especially in forests), seed dispersal (animals carry larger seeds to suitable sites, faeces as fertiliser, gut passage stimulates germination) and habitat provision (trees as nest sites and canopy microclimate; mollusc shells for hermit crabs) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **evapotranspiration** → evaporation plus transpiration • **humus** → dark, decomposed organic matter in soil • **reliable** → repeatable / gives consistent results when repeated **Do not credit** • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	6

Do not write outside the box

Question	Marking guidance	Marks
02.2	**1 mark each** 1 mark for each of the 2 points below. **1** Its nectar is at the end of a 30 cm tube reached only by the long tongue of the sphinx moth, which carries pollen only between orchids of the same species **1 Advantage:** no pollen wasted on other species and the moth has an exclusive food source – risk: if either species declines the other cannot survive **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
02.3	**1 mark each** 1 mark for each of the 2 points below. **1** They are not ruminants so seeds are not destroyed by repeated chewing / pass through intact – they eat the fruit of most tree species **1** Not territorial so they carry seeds over large distances / areas **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 18-19; AQA spec 3.1.2.1.3.	
03.1	**2 marks per point** 2 marks for each of the 3 points below; award 1 mark where a point is only partly made. **2 Selective breeding:** parents chosen for desirable genetic characteristics (Limousin muscle growth, Merino wool, Cheviot hardiness) – combines genes but characteristics of offspring not fully predictable – many generations needed – risk of inbreeding **2** Asexual reproduction (cuttings, runners, tissue culture, cloning): offspring genetically identical to the parent so characteristics are predictable and survival high – but no variation so no improvement and vulnerability to the same disease – cloning of animals (nucleus transfer into an enucleated egg, surrogate) still developing **2 Cross-breeding:** two breeds combined to give a mix of characteristics with hybrid vigour (heterosis) and lower inbreeding risk, eg Zebu (heat tolerant) x Ayrshire (high milk yield) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round.	6

Do not write outside the box

Question	Marking guidance	Marks
03.2	**1 mark each** 1 mark for each of the 4 points below. **1 AI:** semen from a male with desirable characteristics can fertilise many more females than natural mating – frozen semen stored and transported easily and used after the male dies **1 Embryo transfer:** FSH stimulates a superior female to release many eggs, which are fertilised (IVF) with sperm from a superior male **1** Each embryo is implanted in a different surrogate female **1** So the donor produces many offspring per year rather than one per pregnancy **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 321-323; AQA spec 3.5.1.2.	
04.1	**1 mark each** 1 mark for each of the 4 points below. **1** Natural processes are driven by renewable, low energy-density solar energy captured by photosynthesis – human processes usually use high energy-density non-renewable fossil fuels **1** Natural processes occur at low temperatures because enzymes lower the activation energy of reactions – human processes need high temperatures and pressures **1** Natural processes produce no toxic wastes / their products are raw materials for other processes – industrial processes emit CO_2 and NOx **1** So natural fixation is sustainable indefinitely with a low carbon footprint **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	4

Do not write outside the box

Question	Marking guidance	Marks
04.2	**Any 3 of 5** 1 mark each for any 3 of the 5 points below, in any order, maximum 3. **1** Hydrological cycle (evaporation, precipitation) **1** Carbon cycle / photosynthesis **1** Atmospheric circulation / winds **1** Thermohaline circulation (density differences from heating / evaporation) **1** Decomposition **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Thermohaline** → the ocean conveyor driven by temperature and salinity **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'plants breathe in carbon dioxide and breathe out oxygen' - name photosynthesis and respiration	3
04.3	**1 mark each** 1 mark for each of the 3 points below. **1** Bioleaching and phytomining extract metals at low temperature instead of high-temperature smelting **1** Anaerobic digestion and composting treat wastes – enzymes in industry (biological washing powders, biofuels from algae / microorganisms) **1** Heat pumps and solar thermal capture low-density heat – low-temperature manufacturing using biological polymers – fuel cells instead of combustion **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373; AQA spec 3.6.2.	
05.1	**Marks shared** Up to 4 marks in total - share them between the 3 points below according to how completely each one is made. **up to 2 Extensive:** low inputs of energy, food and chemicals so low cost and low environmental impact – sustainable – but low yield per area and dependent on natural productivity **up to 2 Intensive:** control of all limiting factors gives very high yield and profit per area – but high inputs (fishmeal from wild fish, energy), disease and lice at high density, pollution from wastes, antibiotic / pesticide use, escapes **up to 2** Judgement depends on species, market and site **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **aquaculture** → fish or shellfish farming • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	4

Do not write outside the box

Question	Marking guidance	Marks
05.2	**1 mark each** 1 mark for each of the 3 points below. **1** Adults with desirable characteristics (fast growth, shape, colour) stripped of eggs and milt which are mixed for fertilisation **1** Eggs raised in aerated freshwater tanks, dead / diseased eggs removed daily – fry fed fishmeal pellets and moved through tanks as they grow **1** After 12-18 months smolts transferred to seawater cages, fed pellets until harvest at 3-5 kg **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
05.3	**1 mark each** 1 mark for each of the 3 points below. **1** About 10% of the world's mangroves have been cleared for coastal shrimp lagoons (up to 100 ha), removing nurseries and coastal protection **1** Lagoons need water exchange to remove wastes and aeration to prevent deoxygenation as wastes decompose **1** Eyestalk ablation stimulates breeding – high densities increase disease so antibiotics / pesticides used **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **deoxygenation** → loss of dissolved oxygen from the water **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 352-356; AQA spec 3.5.2.3.	
06.1	**1 mark each** 1 mark for each of the 2 points below. **1** Its reading is adjusted to match a meter known to be accurate so that the results are accurate / comparable **1** If it cannot be adjusted the percentage error should be recorded so the results can be corrected **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **accurate** → close to the true value	2

Do not write outside the box

Question	Marking guidance	Marks
06.2	**Any 3 of 6** 1 mark each for any 3 of the 6 points below, in any order, maximum 3. **1 Droppings:** diet, gender, territory size **1 Owl pellets:** diet **1** Feeding marks (eg chewed hazel nuts): presence of species such as dormice **1 Nests / burrows:** presence and breeding **1 Tracks / footprints:** presence, movements **1** Territorial marks (scratching posts): presence and territory **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
06.3	**1 mark each** 1 mark for each of the 2 points below. **1** Bat detectors / sonograms record the ultrasonic sounds bats produce, which differ between species **1** The recordings provide evidence of presence, abundance and activity of species that cannot easily be seen **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
06.4	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Marked individuals mix freely with the rest of the population **1** Marks do not fall off or wear away, and marking does not affect survival **1** The population does not change during the study (no births, deaths or migration) **1** All individuals have an equal chance of being caught / animals are mobile and can be caught **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **mark-release-recapture** → estimating population size from the proportion of marked animals recaptured **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 402, 405, 407; AQA spec 3.1.2.3.5-3.1.2.3.6.	
07.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1 A:** The readings are precise but may not be accurate **Also accept** • **precise** → readings close together / fine resolution • **accurate** → close to the true value	1

Do not write outside the box

Question	Marking guidance	Marks
07.2	**1 mark each** 1 mark for each of the 2 points below. **1** A result that differs from other results with which it was expected to be similar **1 Repeat the study / measurement:** if it recurs it is a 'real' result caused by another variable – if not it was an error and can be excluded **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **anomalous** → an outlier / a result that does not fit the pattern **Do not credit** • discarding a result only because it does not fit the expected pattern, with no reason to think it is faulty	2
07.3	**1 mark each** 1 mark for each of the 2 points below. **1** To test the reliability of the method / equipment **1** To decide the sample size, number of samples, spacing or timing needed to detect the variation without collecting unnecessary data **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **reliability** → repeatable / gives consistent results when repeated **Do not credit** • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 390-391; AQA spec 3.7.1.	
08.1	**1 mark each** 1 mark for each of the 3 points below. **1 South America:** high loss (5.1-5.5 Mha yr^{-1}) falling by about half by 2020 **1 Africa:** loss steady at about 3.4 then rising to 3.9 by 2020 - now the highest **1 Europe:** negative values - forest area increasing (afforestation exceeds loss), though the gain is shrinking **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3

Do not write outside the box

Question	Marking guidance	Marks
08.2	**Marks shared** Up to 4 marks in total - share them between the 3 points below according to how completely each one is made. **up to 2 South America:** clearance for commercial cash crops (sugar cane, soya) and cattle ranching for export – HEP reservoirs – roads improving access – mining **up to 2 Africa:** growing rural populations clearing land for subsistence farming – fuelwood and charcoal collection above regrowth (charcoal wastes much of the wood's energy) – fodder cutting in dry areas **up to 2 Both:** hardwood timber (teak, mahogany) from clear-felled mixed forest often not replanted – urban expansion **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **HEP** → electricity generated from falling water **Do not credit** • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	4
08.3	**1 mark each** 1 mark for each of the 2 points below. **1** They create barriers fragmenting wildlife populations **1** Improved access makes it easier to remove timber and farm produce, so clearance spreads along them **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
08.4	1 mark for the point below. **1** Conifers grow quickly so plantations are replanted after harvest and yield a saleable crop within decades – slow-growing hardwoods (teak, mahogany) take too long for commercial plantations so they are taken from natural forest that is clear-felled and not replanted **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 367-368; AQA spec 3.5.3.3.	
09.1	**1 mark each** 1 mark for each of the 3 points below. **1** The nucleus of a stem cell from the endangered species is transferred into an empty egg of a closely related species (southern white rhino) **1** The embryo is implanted into a surrogate female of the related species which gives birth to a northern white rhino **1** Very low success rates with wildlife so far – deep-frozen cells could allow clones even after the last individuals die **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3

Do not write outside the box

Question	Marking guidance	Marks
09.2	**1 mark each** 1 mark for each of the 3 points below. **1 Micro-propagation:** tissue culture producing many cell clusters from one plant, each grown into a genetically identical plant **1 Black poplar:** cuttings / pollarded stems of pure native trees are rooted to produce many identical trees **1** Needed because the flowers are pollinated by introduced subspecies producing hybrids, so cloning maintains the pure gene pool (but no genetic diversity) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **gene pool** → the range of alleles / genes present in a population	3
09.3	**1 mark each** 1 mark for each of the 4 points below. **1** Stores seeds of wild plants (especially threatened species and crop wild relatives) so species are not lost if extinct in the wild **1** Seeds collected worldwide – target of 10 000 seeds per species to maintain a diverse gene pool **1** Stored dry and refrigerated / frozen in an underground reinforced vault to reduce risk of accident **1** Seeds germinated periodically to check viability and replenish stocks – can be used for reintroduction / breeding **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **gene pool** → the range of alleles / genes present in a population	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 35-37; AQA spec 3.1.2.3.3.	

Do not write outside the box

Question	Marking guidance	Marks
10.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Why needed:** growing population (to 12 billion), land degradation since the fertile crescent, reliance on artificial fertilisers, pesticides and unsustainable irrigation – limits to new land and imports • **Pest control:** cultural methods (weeding, mulching, rotation, barrier / companion crops, predator habitats, biological control, sterile males, pheromones), resistant varieties, integrated control, reduced antibiotics • **Nutrients:** natural processes - nitrogen-fixing legumes, decomposition, recycling organic matter (manure, compost), crop rotation, permaculture, conservation of soil biota and mycorrhizae, reduced artificial fertiliser – phosphate recycling • **Energy:** natural processes instead of Haber-process fertiliser, low tillage, less machinery, renewable energy, low food miles, reduced methane (cattle diets, rice varieties) • **Water:** low water-use crops, drip irrigation, reservoirs and aquifer recharge, avoiding saline water – soil: organic matter maintenance, erosion control • **Biodiversity:** retaining hedgerows, ditches, ponds, woodlands – CWR conservation and seed banks – agri-environment schemes • **Social:** control of impacts on rural communities, Fairtrade, local food – consumer choices • **Evaluation:** yields vs sustainability, cost, need for policy support – judgement **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content.	9

Do not write outside the box

Question	Marking guidance	Marks
	• **sustainability** → meeting present needs without stopping future generations meeting theirs • **biological control** → using a natural predator, parasite or pathogen of the pest • **nitrogen-fixing** → nitrogen gas converted into ammonium or other nitrogen compounds **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	
	Model answer - Level 3 (8 marks)	
	Sustainability matters because the population may reach 12 billion, land has been degraded since farming began in the fertile crescent, and current yields depend on artificial fertilisers, pesticides and irrigation that draws down groundwater, with little new land to cultivate. Pest control can move from chemicals to cultural methods - weeding, mulching, rotation, barrier and companion crops, habitats for predators - biological control, sterile males and pheromones, resistant varieties and integrated control, with fewer antibiotics in livestock.	
	Nutrients can come from natural processes: legumes fixing nitrogen, recycling manure and compost, rotation, permaculture and conserving soil biota and mycorrhizae, reducing manufactured nitrate and the non-renewable rock phosphate it accompanies. Energy use falls with low tillage, less machinery, renewable power, low food miles and lower methane from cattle diets and drought-tolerant rice. Water is conserved with low water-use crops, drip irrigation, reservoirs and aquifer recharge and by avoiding saline water; soil by maintaining organic matter and controlling erosion. Retaining hedgerows, ditches, ponds and woodland, conserving crop wild relatives in seed banks and agri-environment schemes protect the biodiversity that supplies pollination and pest control, while Fairtrade, local food and consumer choices address the social side.	
	The tension is between yield and sustainability: some strategies cost more or yield less in the short term and need policy support to be adopted. But since the unsustainable inputs themselves depend on finite fossil fuels and phosphate, the strategies that use natural processes are the only ones that can feed a larger population indefinitely.	
	Weaker answer - Level 2 (4 marks)	
	Agriculture can be made more sustainable by using less pesticides and using biological control instead, using manure instead of artificial fertilisers, growing legumes which add nitrogen to the soil, and using crop rotation. Drip irrigation saves water. Farmers can keep hedgerows to provide habitats for wildlife. Organic farming uses these methods. These strategies reduce pollution and keep the soil healthy so food can be produced in the future.	
	What makes the difference	
	• Level 2 lists valid strategies but does not explain why they are needed (population growth, land degradation, dependence on finite inputs) or group them by the problem they solve.	
	• Level 3 covers energy, soil, biodiversity and social strategies as well as pests, nutrients and water.	

Do not write outside the box

Question	Marking guidance	Marks
	• Level 3 discusses the yield-sustainability tension and the need for policy support; Level 2 has no evaluation.	
10.2	1 mark for the point below. **1** Agricultural systems designed to mimic natural ecosystems - perennial crops, mixed species, recycling of nutrients, minimal external inputs **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **recycling** → reprocessing a material to make something new **Do not credit** • 'recycling uses no energy' - collection, sorting and reprocessing all use energy, usually less than making the material from new	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 335-337; AQA spec 3.5.1.5.	

Do not write outside the box

Question	Marking guidance	Marks
11.1	**Using named examples, discuss how the ecological features of a habitat determine its importance, the threats it faces and the conservation strategies that are appropriate.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Temperate broadleaf woodland:** seasonality, deep soils, high biodiversity, resources, climate and soil regulation – deforestation, fragmentation, management change – SSSIs, ancient woodland protection, coppicing, mixed planting • **Tropical rainforest:** stable warm wet climate, stratification, very high biodiversity, carbon and hydrological roles, thin soils – fuelwood, timber, agriculture, mining, reservoirs, climate change – protected areas, debt-for-nature swaps, sustainable exploitation • **Tropical coral reefs:** cnidarian-zooxanthellae symbiosis, light, temperature, turbidity, salinity – fisheries, erosion protection, medicines, tourism – damage, souvenirs, sedimentation, pollution, bleaching, fishing, introduced species – MPAs, no-take zones, control of activities • **Deep-water reefs:** cold, dark, slow growth – research, fisheries – trawling, oil / gas, acidification – SACs • **Oceanic islands:** isolation, endemism, no mammal predators	25

Do not write outside the box

Question	Marking guidance	Marks
	– exploitation, introduced species, habitat change, sea level – eradications, control of development and visitors • **Mangroves:** halophytes, anaerobic mud – erosion protection, fisheries nurseries, sediment trapping – clearance for aquaculture / urban / ports, pollution, climate change – protection, replanting • **Antarctica:** extreme cold, seasonality, marine nutrients, ice – water store, albedo, carbon, research – climate change, ozone, tourism, fishing, minerals – Antarctic Treaty, CCAMLR, tourism and waste rules • **Synthesis:** features -> importance -> threats -> strategy – evaluation of success – conclusion *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 50-71; AQA spec 3.1.2.3.4.	

Do not write outside the box

Question	Marking guidance	Marks
11.2	**Discuss how human activities influence biodiversity, using examples from a range of different contexts.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Direct exploitation:** food (fisheries, bushmeat), fashion, entertainment / pets, furniture and ornaments, traditional medicines, other products (whale oil) – deliberate eradication of predators, competitors, vectors, pests • **Changes in abiotic factors:** water availability (drainage, abstraction, HEP), light (forest clearance), oxygen (organic pollution, thermal), nutrients (eutrophication), pH (acid mine drainage, acid rain), temperature (thermal pollution, climate change), turbidity, physical damage – inactivity (stopping plagioclimax management) • **Changes in biotic factors:** introduced competitors, predators, pathogens, hybridisation – loss of pollinators, seed dispersers, keystone species – food chain impacts (sea otters, puffins, jellyfish) • **Habitat destruction:** deforestation, farmland expansion, urbanisation, mining, reservoirs – fragmentation and edge effects • **Contexts:** temperate woodland, rainforest, coral reefs, oceanic islands, mangroves, Antarctica	25
	– agriculture, fishing, forestry • **Positive influences:** habitat creation, conservation management, reintroductions, legislation • Evaluation of which influences are most damaging and hardest to reverse – conclusion *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 20-26, 328-330, 341-344, 367-369; AQA spec 3.1.2.2, 3.5.1.3, 3.5.2.2, 3.5.3.3.	

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